

Sorting Quadrilaterals by Their Attributes

Right angles, parallel sides, equal sides, and the groups shapes share

Directions: The six shapes below are the **Shape Set**. Every problem except 5, 8 and 9 uses it. Three attributes matter on this page:

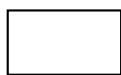
- a **square corner** (a right angle) — the corner of an index card fits it exactly;
- a **pair of parallel sides** — two sides that stay the same distance apart;
- **equal sides** — sides of the same length.

Use a ruler or the corner of a card to check. Answer in whole numbers.

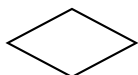
The Shape Set



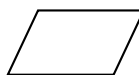
A



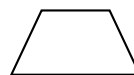
B



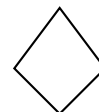
C



D



E



F

1. Look at shape B in the Shape Set. How many square corners does it have?

Answer: _____

2. Look at shape E. How many pairs of parallel sides does it have?

Answer: _____

3. Sort the whole Shape Set. How many of the six shapes have four equal sides?

Answer: _____

4. Shape A and shape E are both quadrilaterals, but they do not share every attribute. How many more pairs of parallel sides does shape A have than shape E?

Answer: _____

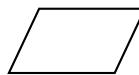
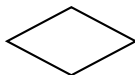
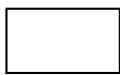
5. A tile shop sold quadrilateral floor tiles last week: 9 square tiles, 12 rectangle tiles, 4 rhombus tiles and 5 trapezoid tiles. All four kinds are quadrilaterals — they all have four straight sides. How many tiles did the shop sell in all?

Answer: _____ tiles

6. Count the pairs of parallel sides on each of the six shapes in the Shape Set. Most of the shapes share the same count. What is that shared number?

Answer: _____

7. Below are a square, a rectangle, a rhombus and a parallelogram.



- (a) Circle every shape above that has four square corners.
- (b) Explain why the square belongs in that group but the rhombus does not, even though both of them have four equal sides.

8. At the tile shop, the tiles with four square corners are the square tiles and the rectangle tiles. Using last week's sales (9 square tiles, 12 rectangle tiles, 4 rhombus tiles, 5 trapezoid tiles), how many tiles sold had four square corners?

Answer: _____ tiles

9. Still at the tile shop.

- (a) A trapezoid has exactly one pair of parallel sides. How many trapezoid tiles were sold?

- (b) Explain why the trapezoid tiles cannot join the group of tiles with four square corners.

10. Go back to the Shape Set.

- (a) How many of the six shapes have at least one pair of parallel sides? _____
- (b) Shape A can be put in more than one group at the same time. Explain how that is possible, and name two groups it belongs to.